Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 7 · SLO 2 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 7 : SLO â€“ 2 | Implementation of Linked Lists â€” Concepts

■ I. Fill in the Blanks â€” Linked List Basics

• A linked list is a collection of NODES, where each node contains data and a pointer to the NEXT NODE.
• In C, the keyword STRUCT is used to define a node in a linked list.
• The pointer used to point to the first node of the linked list is called the HEAD (START).
• The pointer in the last node of a singly linked list points to NULL.
• A node in a linked list typically contains two fields: DATA and NEXT (LINK).
• Dynamic memory allocation for a new node is done using the MALLOC() function.

■ II. Fill in the Blanks â€” Traversal, Insertion and Deletion

• To traverse a linked list, a TEMPORARY (PTR) pointer is used to iterate through each node.
• To insert a node at the beginning of the linked list, update the new node's link to point to the CURRENT HEAD (START).
• After inserting a node at the beginning, the HEAD (START) must be updated to point to the new node.
• To delete a node, it's important to FREE the memory after unlinking the node.
• A NULL check should always be done before accessing a node's data through a pointer.
• Inserting a node at the end requires traversal until the node's NEXT pointer is NULL.